LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Practice Pack: Metals and Non-metals

Chapter / Topic: Topic-based: use with whichever Class 8 book your students follow

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PART A: BASICS

Q1. Give four physical properties of metals.

Answer: (Any four) They are lustrous (shiny), hard, malleable (can be beaten into sheets), ductile (can be drawn into wires), good conductors of heat and electricity, and sonorous (they make a ringing sound).

Q2. Name a metal that is a liquid at room temperature, and a non-metal that is a liquid.

Answer: Metal: mercury. Non-metal: bromine.

Q3. Why are bells made of metal and not of wood?

Answer: Metals are sonorous: they make a ringing sound when struck.

Q4. What is the nature of the oxides of metals and of non-metals?

Answer: Metal oxides are basic (they turn red litmus blue). Non-metal oxides are acidic (they turn blue litmus red).

Q5. Why is sodium stored in kerosene?

Answer: It reacts vigorously with oxygen and water in the air and can catch fire.

PART B: THINK AND APPLY

Q6. Write the word equation and the balanced chemical equation for burning magnesium in air. What is the nature of the product?

Answer: Magnesium + oxygen → magnesium oxide. $2Mg + O_2 \rightarrow 2MgO$. Magnesium oxide is basic.

Q7. Write the equation for burning sulphur in air. What happens when the product dissolves in water?

Answer: $S + O_2 \rightarrow SO_2$. Sulphur dioxide dissolves in water to form sulphurous acid, which turns blue litmus red (acidic).

Q8. A zinc plate is put in copper sulphate solution. What do you observe? Write the equation.

Answer: The blue colour of the solution fades, and a reddish-brown layer of copper forms on the zinc. Zinc is more reactive than copper and displaces it: $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$.

Q9. Will a reaction take place if copper is put in zinc sulphate solution? Why?

Answer: No. Copper is less reactive than zinc, so it cannot displace zinc.

Q10. Why are copper and aluminium used to make electric wires?

Answer: They are good conductors of electricity and are ductile (can be drawn into thin wires).

Q11. Why do iron objects rust, but gold jewellery stays shiny for years?

Answer: Iron reacts with oxygen and moisture in the air to form rust. Gold is very unreactive and does not react with air or water.